

GDAM GREEN HYDROGEN

Levelised Cost of Green Hydrogen from GDAM

Methodology, Assumptions and Interpretation

Data: IEX GDAM, 15-minute block prices, Apr 2022 – Aug 2026

Scope: Gujarat and Rajasthan | Exchange-based procurement only | No storage

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REVISION 2 Revision 2 adds user-configurable **additional charges and subsidies** (Section 7). Several items previously listed as unmodelled — electricity duty, STU demand charges, land, insurance, working capital, SIGHT capital support, GST reimbursement — can now be entered directly. They remain excluded from the defaults, so all figures quoted in this document are unchanged.

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Executive Summary

This document describes the GDAM Green Hydrogen LCOH Explorer, an open-source interactive dashboard for green hydrogen (GH2) project developers in India. Its purpose is to help a developer determine the levelised cost of hydrogen (LCOH) from an alkaline electrolyser procuring power exclusively from the IEX Green Day-Ahead Market (GDAM), with no captive renewable plant and no battery storage.

The tool ingests four years of real 15-minute GDAM block prices (Aug 2022 – Aug 2026) and computes LCOH as a function of a single operational choice: the maximum landed power price the developer is willing to pay. Everything else follows from the price distribution and the cost stack.

Key finding	Value
Best available LCOH (Gujarat, AY2025-26, 20-yr)	Rs 268/kg
Best available LCOH (Gujarat, AY2025-26, 7-yr, 0% residual)	Rs 309/kg
Best available LCOH (Rajasthan, AY2025-26, 20-yr)	Rs 278/kg
Grey hydrogen benchmark	Rs 250/kg
Optimal capacity factor (all years)	64 – 69%
Optimal bid ceiling (Gujarat, AY2025-26)	Rs 4.98/unit landed (Rs 4.43 MCP)
Required plant size for 500 TPA at optimum	4.60 MW
Stack replacement year (at optimum)	Year 10.6
Rajasthan cost penalty vs Gujarat	Rs 10–12/kg (charge structure only)

IMPORTANT This tool is a first-pass scoping instrument for the GH2 producer. It uses a single representative year of exchange prices as a proxy for long-run power cost. The low-price regime observed from 2024 onward is approximately 2.5 years old. Costs excluded from the defaults — electricity duty, STU demand charges, GST, land, insurance, working capital — are material; they are listed in Section 13 and can be entered through the additional-charges fields described in Section 7.

1. What This Tool Does

An alkaline electrolyser is connected to the GDAM. No captive solar or wind plant, no battery. The developer bids into the exchange every day and buys power only when the 15-minute block price is at or below a chosen ceiling. The electrolyser runs at full load whenever a qualifying block is bought, subject to a minimum continuous run to protect equipment (typically 2 hours for alkaline).

For any given ceiling, the historical price data yields an annual operating-hour count and an average power cost across the blocks bought. These, combined with the developer's cost stack, produce the LCOH. The dashboard sweeps the ceiling from the year's minimum price to its maximum and plots the resulting LCOH curve. The curve has a characteristic U-shape: too low a ceiling means too few hours, driving up unit fixed costs; too high a ceiling means buying expensive evening peak power, driving up the power component.

The developer's decision variable is the ceiling. Everything else is an output.

2. Data

2.1 Source

IEX Green Day-Ahead Market (GDAM), 15-minute block prices. The raw file covers 1 April 2022 to 21 August 2026, 153,984 blocks. Two fields are spliced: MCP (Rs/MWh) is used to 20 February 2024, and unconstrained_m_c_p from 21 February 2024 onward. Both are converted to Rs/kWh by dividing by 1,000. Unconstrained MCP is used throughout; 679 of 87,648 flagged blocks (0.77%) were congested — too small a share to move an average, and excluded consistently.

2.2 Price years

Four complete August-to-August windows are used. Calendar years are deliberately avoided because they truncate the seasonal cycle: the Jan–Mar dead zone (zero hours below Rs 1.5 MCP) and the Sep–Oct cheap window are the two most important seasonal features, and a calendar year boundary splits at least one of them.

Year	Window	Days	Mean	p10	p25	Median
AY2022-23	22 Aug 22 – 21 Aug 23	364	5.80	3.40	3.93	4.89
AY2023-24	22 Aug 23 – 21 Aug 24	365	5.73	3.05	3.89	4.54
AY2024-25	22 Aug 24 – 21 Aug 25	364	4.61	2.10	3.00	4.00
AY2025-26	22 Aug 25 – 21 Aug 26	364	4.61	1.63	2.60	3.67

Year	h below Rs 1.5	h below Rs 2.5	h below Rs 3.5
AY2022-23	0	43	1,030
AY2023-24	23	359	1,366
AY2024-25	338	1,321	3,248
AY2025-26	676	2,059	3,761

Table 1: GDAM price statistics by Aug-Aug year. All prices in Rs/kWh; h values in hours per year.

AY2022-23 and AY2023-24 show near-zero hours below Rs 1.5 MCP. Any LCOH result from these years should be read as a historical benchmark, not a forward projection. The low-price regime — driven by growing solar surplus — is a post-2024 phenomenon.

3. Landed Cost Formula by State

Open-access GH2 producers qualify for concessions under the Ministry of Power's Green Hydrogen Open Access notification. The landed cost is exchange MCP plus residual transmission and wheeling charges, times a composite loss factor:

State	Formula	Adder (Rs/unit)	Loss factor	Basis
Gujarat	$\text{landed} = (\text{MCP} + 0.250) \times 1.065$	0.250	1.065	50% wheeling waiver; ISTS waived
Rajasthan	$\text{landed} = (\text{MCP} + 0.365) \times 1.088$	0.365	1.088	50% wheeling waiver; ISTS waived

Table 2: State landed-cost formulas.

The Rs 10–12/kg Rajasthan cost penalty relative to Gujarat is entirely attributable to the higher residual charge stack. The exchange price is identical for both states — GDAM is a single national market clearing at one price per block.

IMPORTANT The ISTS waiver applies only to projects commissioned before 31 December 2030. Electricity duty and STU demand charges are NOT included in these formulas and are excluded from the default LCOH calculation. They can be entered through the additional-charges fields (Section 7). See also Section 13.

Why the sweep is in MCP space

Both formulas are monotonic affine transforms of MCP ($\text{landed} = a \cdot \text{MCP} + b$, where a and b are positive constants). Affine monotonic maps preserve both ordering and arithmetic means. Consequently: (i) the set of blocks selected at a landed ceiling C equals the set selected at the MCP threshold $t = C/a - b/a$; (ii) the mean landed cost of selected blocks is $a \cdot w + b$, where w is the mean MCP of those blocks. The browser therefore looks up the pre-computed MCP sweep and applies the state transform client-side. One sweep serves every state.

4. Dispatch Logic — the Min-Run Rule

GDAM is a day-ahead market. Block prices are published the previous evening, so the dispatch schedule is executable without forecast error. No DSM penalties or imbalance charges arise from GDAM trading.

Pure on/off dispatch at block level does not suit alkaline electrolyzers: cold-start requires a thermal ramp, and sub-15-minute shutdowns accumulate cell degradation. The tool imposes a minimum continuous run (default 2 hours = 8 blocks):

Step	Rule
1	Qualify blocks: all blocks with landed price at or below the ceiling are candidates.
2	Identify contiguous runs of qualified blocks.
3	Runs of at least min_run blocks pass through unchanged.
4	Shorter runs are extended outward, one block at a time, always taking the cheaper of the two adjacent blocks, until the run reaches min_run blocks.
5	The extended window is kept ONLY if its mean landed price is at or below the ceiling. Otherwise the entire window is discarded.

The penalty at the optimum is Rs 0.0–0.4/kg. At the optimal 64–69% capacity factor the electrolyser runs two-thirds of all blocks. Cheap periods are multi-hour solar-surplus windows and overnight troughs, not scattered single blocks. The minimum-run filter therefore almost never binds at the LCOH optimum. It would bind hard on a very low ceiling strategy (below 5% CF), providing an additional argument against such designs.

5. LCOH Formula — Full Derivation

Step 1 — Lookup operating hours and mean MCP

```
t      = ceiling_landed / loss_factor - adder
hours,w = interpolate(sweep, t)      <- linear, never extrapolated
landed  = (w + adder) x loss_factor
```

Step 2 — Production

```
annual_kg = MW x 1000 x hours x load_factor / SEC
CF         = hours / 8760
```

Step 3 — Capital cost annualisation

```
capex_gross = MW x 1000 x (capex_per_kw + additional_capex_per_kw)
capex_net   = max(capex_gross - MW x 1000 x capex_subsidy_per_kw, 0)
CRF(r,N)    =  $r(1+r)^N / ((1+r)^N - 1)$ 
ann_capex   = (capex_net - capex_net x residual / (1+r)^N) x CRF(r,N)
annual_OM   = capex_gross x om_pct
```

Step 4 — Stack replacement sinking fund

```
stack_cost = MW x 1000 x ELY_kw x stack_pct
interval    = stack_life_hours / hours
sinking     = CRF(r,N) x SUM stack_cost / (1+r)^(k x interval)
              for k = 1,2,3,... while k x interval < N
```

Step 5 — Total LCOH

```
LCOH = (ann_capex + annual_OM + sinking) / annual_kg
      + SEC x landed
      + water_cost_per_kg
      + additional_opex_per_kg      <- Section 7
      - production_subsidy_per_kg   <- Section 7
      - power_subsidy_per_unit x SEC <- Section 7
```

With the Section 7 fields at their zero defaults, this reduces exactly to the original formula. All reference figures in this document are computed at those defaults.

Reference case — Gujarat, AY2025-26, 20-year recovery, no additional charges or subsidies:

Component	Calculation	Rs/kg	% of LCOH
Capital recovery	$285,183,990 \times 0.117460 \div 500,000$	67.00	25.0%
Fixed O&M	$285,183,990 \times 0.04 \div 500,000$	22.81	8.5%
Stack replacement	$PV = 26,093,399; \times 0.117460 \div 500,000$	6.13	2.3%
Power (landed)	$52 \times (2.843 + 0.250) \times 1.065$	171.29	63.9%

Component	Calculation	Rs/kg	% of LCOH
Water	$15 \text{ L/kg} \times \text{Rs } 60/\text{kL} \div 1,000$	0.90	0.3%
LCOH Total		268.13	100.0%

Table 3: LCOH breakdown. Design: 4.599 MW, 5,654 h/yr, CF 64.5%, capex Rs 28.52 cr. $CRF(10\%,20) = 0.117460$.

6. Capex Build-Up

All uplifts are applied to the base electrolyser installed cost. The same Rs 62,010/kW total is the default across both states; the developer can override any line item in the Advanced panel.

Item	Uplift	Rs/kW	Running total
Electrolyser installed (equipment + BOP + installation)	—	39,000	39,000
Civil works / site prep	12%	4,680	43,680
Water treatment (DM/RO)	6%	2,340	46,020
Power evacuation, HT yard, transformer	10%	3,900	49,920
H2 purification, drying, compression	10%	3,900	53,820
Instrumentation, safety, fire, controls	6%	2,340	56,160
Owner cost, EPC margin, contingency	15%	5,850	62,010
All-in plant capex	+59%	62,010	62,010

Table 4: Capex build-up. All uplifts are on the base electrolyser installed cost of Rs 39,000/kW.

IMPORTANT These are indicative uplifts based on Indian EPC norms for MW-scale projects. They have NOT been validated against a project-specific bill of quantities. A real BOQ could move the all-in figure by $\pm 20\%$. Since capex recovery is 25–40% of total LCOH at the optimum, a 20% capex reduction saves roughly Rs 15–20/kg — enough to reach grey parity from the AY2025-26 base without any other intervention.

7. Additional Charges and Subsidies

The default cost stack deliberately excludes a number of real costs, and excludes all government support other than the transmission and wheeling waivers already embedded in the landed-cost formula. Section 13 lists those exclusions. This section describes the mechanism by which a developer can enter them.

Every field described here defaults to zero. At the defaults, the model is numerically identical to the version without them, and every figure quoted elsewhere in this document remains exact. The exclusions in Section 13 are therefore defaults, not hard limits of the tool.

7.1 Additional charges

Two inputs, in the basic panel. The capex field is entered in Rs/kW so that it scales with plant size in the same way as the rest of the capital stack. The opex field carries a unit selector, because the excluded items are naturally quoted in different bases: electricity duty in Rs per unit of power, insurance as a percentage of capex, land lease as a lump sum per year.

Field	Unit	Typical use
Additional capex	Rs/kW	Land acquisition, site infrastructure, items absent from the BOP uplifts, connection deposits
Additional opex	selectable — see 7.2	Electricity duty, STU demand charges, insurance, working capital carry, statutory fees

Table 5: Additional charge inputs.

7.2 Unit conversions for additional opex

The entered value is converted to Rs/kg before it enters the LCOH. The conversions are:

Selected unit	Conversion to Rs/kg	Worked example at the reference case
Rs/kg H2	value	Rs 10.00/kg → +Rs 10.00/kg
Rs/unit power	value × SEC	Rs 0.10/unit × 52 → +Rs 5.20/kg
Rs/MW/year	value × MW ÷ annual_kg	Rs 10 lakh × 4.599 ÷ 500,000 → +Rs 9.20/kg
Rs/year	value ÷ annual_kg	Rs 50 lakh ÷ 500,000 → +Rs 10.00/kg
% of capex/year	pct × capex_gross ÷ annual_kg	0.4% × Rs 28.52 cr ÷ 500,000 → +Rs 2.28/kg

Table 6: Additional-opex unit conversions. Reference case: 4.599 MW, 500 TPA, SEC 52 kWh/kg, capex Rs 28.52 cr.

The Rs/unit power conversion multiplies by SEC, which is equivalent to multiplying by annual units drawn divided by annual kg produced. This is the correct treatment for a per-unit charge such as electricity duty: it applies only to energy actually purchased, and therefore only during operating hours. Nameplate hours are not used anywhere in this conversion.

7.3 Subsidies

Three independent fields in the Advanced panel, all optional and all simultaneously applicable. A developer claiming SIGHT capital support alongside a GST reimbursement would populate two of them.

Field	Unit	Enters the model as	Typical use
Capital subsidy	Rs/kW	Reduction to capex at year zero, before CRF	SIGHT Component II, state capital subsidy

Field	Unit	Enters the model as	Typical use
Production subsidy	Rs/kg H ₂	Direct deduction from LCOH	GST reimbursement, per-kg production incentive
Power subsidy	Rs/unit power	Deduction of value × SEC from LCOH	State power tariff rebate, duty exemption

Table 7: Subsidy inputs. GST reimbursement is entered through the production subsidy field, since it is economically an offset per unit of output.

7.4 Treatment decisions

Five choices materially affect the result. They are documented here because a user reconciling this model against their own spreadsheet will otherwise find unexplained differences.

#	Decision	Rationale
1	Fixed O&M is charged on gross capex — including additional capex, and not reduced by a capital subsidy.	The physical asset requires maintenance regardless of who funded it. A capital grant changes the financing, not the maintenance burden.
2	Additional capex attracts O&M at the same percentage as base capex.	A single rate is applied for now. If the additional capex is predominantly land, which carries little maintenance, this overstates O&M — adjust the O&M percentage to compensate.
3	A capital subsidy is front-loaded — deducted at year zero, before the CRF is applied.	Avoids modelling disbursement timing, which varies by scheme and is not reliably predictable. A subsidy received in a later year is worth less than this treatment implies.
4	Residual value is a percentage of net capex.	Follows the Ind AS 20 convention under which a capital grant reduces the carrying amount of the asset. Also prevents a large subsidy combined with a high residual percentage from producing a negative capital recovery charge.
5	Stack cost is unaffected by additional capex or by any subsidy.	Stack cost is derived from the base electrolyser cost only. A subsidy on the initial build does not recur at replacement.

Table 8: Treatment decisions for additional charges and subsidies.

7.5 Effect on MW-invariance

Section 10 establishes that LCOH is mathematically independent of plant size. That property survives every field in this section but one.

Additional capex (Rs/kW), the capital subsidy (Rs/kW) and the Rs/MW/year opex unit are all proportional to MW, so they cancel in exactly the same way as the base capital stack. The Rs/kg and Rs/unit units carry no MW term at all. The percentage-of-capex unit is proportional to capex and therefore to MW.

NOTE The **Rs/year lump-sum** opex unit is the exception. A fixed annual cost spread over more production yields a lower per-kg figure, so LCOH falls as MW rises. This is correct behaviour, not a modelling error — but it means that with a lump-sum cost entered, plant size is no longer a free variable and the optimal ceiling shifts slightly with MW. The dashboard flags this on screen when the unit is selected.

7.6 Worked verification cases

Each case starts from the reference design: AY2025-26, Gujarat, ceiling Rs 4.98, 4.599 MW, 500 TPA, N = 20, r = 10%, residual 0%, base LCOH Rs 268.1/kg. One field is changed at a time.

Case	Change	Mechanism	LCOH
A	Additional opex Rs 10.00/kg	Direct addition	278.1

Case	Change	Mechanism	LCOH
B	Additional opex Rs 0.10/unit	$0.10 \times 52 = +5.20$	273.3
C	Additional opex Rs 50 lakh/yr	$5,000,000 \div 500,000 = +10.00$	278.1
D	Additional opex Rs 10 lakh/MW/yr	$1,000,000 \times 4.599 \div 500,000 = +9.20$	277.3
E	Production subsidy Rs 50/kg	Direct deduction	218.1
F	Power subsidy Rs 0.50/unit	$0.50 \times 52 = -26.00$	242.1
G	Capital subsidy Rs 10,000/kW	Capital recovery 67.00 $\rightarrow$ 56.19; O&M unchanged	257.3
H	Additional capex Rs 5,000/kW	Capital recovery 67.00 $\rightarrow$ 72.39; O&M 22.81 $\rightarrow$ 24.65	275.4

Table 9: Verification cases for additional charges and subsidies. All computed at the reference design.

Case G in full: the subsidy is $4,598.51 \text{ kW} \times \text{Rs } 10,000 = \text{Rs } 4.5985 \text{ cr}$, giving net capex of $\text{Rs } 28.5154 - 4.5985 = \text{Rs } 23.9169 \text{ cr}$. Capital recovery becomes $239,169,000 \times 0.117460 \div 500,000 = \text{Rs } 56.19/\text{kg}$. Fixed O&M remains Rs 22.81/kg, because it is charged on gross capex under decision 1 above. The net effect is $-\text{Rs } 10.80/\text{kg}$.

Case H in full: additional capex is $4,598.51 \text{ kW} \times \text{Rs } 5,000 = \text{Rs } 2.2993 \text{ cr}$, giving gross capex of Rs 30.8147 cr with no subsidy, so net equals gross. Capital recovery becomes Rs 72.39/kg (+Rs 5.40) and O&M becomes Rs 24.65/kg (+Rs 1.84). The net effect is +Rs 7.24/kg. Note that the O&M increase is roughly a third of the capital recovery increase — additional capex is more expensive than it first appears.

The model is exactly additive in these terms. Applying several fields simultaneously produces the base LCOH adjusted by the sum of the individual effects, with one qualification: a capital subsidy and additional capex both act on the same capital chain, so their combined effect is computed on the netted balance rather than as two independent deltas.

8. Reference Results by Price Year

Gujarat GH2 open access. 20-year capital recovery at 10% discount rate. Minimum run 2 hours. Default cost stack, no additional charges or subsidies. AY2025-26 is the recommended base case — the only complete Aug-to-Aug year ending on the most recent data.

Year	Opt. ceiling (landed)	Opt. ceiling (MCP)	Mean landed	Hours	CF	MW (500 TPA)	Capex (cr)	LCOH
AY2022-23	6.39	5.95	4.76	5,756	65.7%	4.52	28.05	342.8
AY2023-24	6.18	5.75	4.50	5,660	64.6%	4.59	28.49	330.7
AY2024-25	5.32	4.75	3.71	6,000	68.5%	4.33	26.89	284.8
AY2025-26	4.98	4.43	3.29	5,654	64.5%	4.60	28.54	268.1

Table 10: Optima by year — Gujarat. MCP equivalent = ceiling_landed / 1.065 – 0.250.

The key structural finding: the optimal ceiling falls from Rs 6.39 to Rs 4.98 across four years — a 22% drop — while the optimal capacity factor holds at 64–69% throughout, and the plant size stays in a 4.33–4.60 MW band. The correct design rule is to size for approximately 65% CF and set the ceiling at whatever the 65th-percentile landed price of the representative year turns out to be. A fixed numerical ceiling is not portable across years.

For Rajasthan, add Rs 10–12/kg to each LCOH above. The Rs 10.2/kg differential at AY2025-26 decomposes as $(3.490 - 3.294) \times 52 = \text{Rs } 10.2/\text{kg}$ in power cost, with all other components unchanged.

9. Interpreting the U-Curve

The LCOH vs landed ceiling chart is the centrepiece of the dashboard. Understanding its shape prevents two common misinterpretations.

9.1 Why the curve rises at low ceilings

A lower ceiling means fewer qualifying hours. Fixed costs are spread over less production, so unit fixed cost rises steeply. At a Rs 1.5/unit ceiling in AY2025-26 the plant runs only 791 hours (9% CF) and capital recovery alone exceeds Rs 640/kg. The power component is very cheap (Rs 78/kg) but irrelevant — it is dominated by fixed costs.

9.2 Why the curve rises at high ceilings

A higher ceiling includes increasingly expensive evening and night blocks. The average landed power cost rises faster than fixed unit cost falls, because once the plant is running 70–80% of hours, most remaining blocks are expensive (Rs 6–10/unit). Buying these adds cost without proportionally reducing fixed unit cost.

9.3 The flat bottom

The optimum is shallow. In AY2025-26 Gujarat, the LCOH is within Rs 5/kg of its minimum across a Rs 1.5/unit ceiling band (roughly Rs 4.0–5.5). Do not read three significant figures of precision into the optimal ceiling number. The capacity factor at the optimum is the stable quantity; the ceiling is the dial that produces it.

9.4 The two ceiling numbers

Every design has two distinct ceiling values that must not be confused. The **bid ceiling** is the number entered into the GDAM order: the landed price above which you stop buying. At the AY2025-26 Gujarat optimum this is Rs 4.98/unit landed, equal to Rs 4.43/unit MCP. The **mean landed cost** is the outcome that enters the power line of the LCOH: Rs 3.29/unit. The Rs 1.69/unit gap is large because most selected blocks cluster far below the marginal one — the price distribution has a long cheap tail.

Because the electrolyser is either running at full rated load or off, every block that is bought draws the same quantity of power. The blocks therefore contribute equally to that mean, and it is a simple arithmetic average of the landed prices of the selected blocks.

9.5 Seasonality

At the AY2025-26 optimum the plant runs 48–81% utilisation by month. The weakest month is April (48%) — pre-monsoon, solar surplus not yet at maximum. The strongest months are September–November (75–81%) when solar surplus is large and peak demand has subsided. January–March run at 59–71%. There is no dead season at the optimum: the plant produces every month, ranging roughly 27–58 t/month against a 42 t/month average. Offtake agreements should be sized around the April minimum, not the January dead zone that applies only to ultra-low ceiling strategies.

10. The MW-Invariance Property

The LCOH-optimal ceiling, optimal operating hours, and capacity factor are mathematically independent of plant size. This is not an approximation — it is an exact algebraic consequence of the cost structure.

```
annual_kg = MW x 1000 x hours / SEC          <- proportional to MW
capex      = MW x 1000 x capex_kw             <- proportional to MW

ann_capex / annual_kg = capex_kw x CRF x SEC / (1000 x hours)  MW cancels
OM / annual_kg        = capex_kw x om_pct x SEC / (1000 x hours) MW cancels
sinking / annual_kg    = ELY_kw x stack_pct x ... / hours      MW cancels
power term             = SEC x landed                      no MW

=> LCOH depends only on hours and landed, not on MW.
```

Practical implication. The developer can independently choose plant size, to hit a target annual production, and operating strategy, to hit a target LCOH. The dashboard shows a single U-curve that applies at any scale. MW scales nameplate capacity, actual output and capex linearly, but does not move the optimal point on the curve.

The property extends to every additional charge and subsidy in Section 7 except the lump-sum Rs/year opex unit, for the reasons given in 7.5.

11. Three-Year Repricing

When the 3-year repricing checkbox is enabled, the dashboard computes a regulatory-style cost-of-service tariff with a declining rate base, allowing the GH2 price to be renegotiated every three years as capital is progressively recovered.

11.1 Mechanics

```
opening rate base at year 1 = capex_net
Year t:
  if t = ceil(stack_life / hours):
    rate_base += stack_cost
    stack_dep = stack_cost / (plant_life - t + 1)
  depreciation = capex_net / N          (while t <= N, else 0)
  return       = r x opening_rate_base
  capital_Rs_kg = (dep + stack_dep + return) / annual_kg
price_b = CRF(r, block_len) x SUM capital_Rs_kg_t / (1+r)^i
          + SEC x landed + OM_per_kg + water
          + additional_opex_per_kg - subsidies_per_kg
```

The rate base opens at net capex, so a capital subsidy reduces the balance on which the developer earns a return. Additional capex does enter the rate base, being capital actually deployed. Additional opex and subsidies are flat across all blocks under the constant-price-year assumption, exactly as the power and water lines are; they shift every block price by the same amount and do not disturb NPV-neutrality.

11.2 Block schedule

Gujarat, AY2025-26, 7-year amortisation, 500 TPA, no additional charges or subsidies.

Block	Years	Opening rate base	Capital charge	Contract price	Revenue/yr	vs grey
B1	1–3	Rs 28.52 cr	130.9	Rs 326	Rs 16.29 cr	1.30×
B2	4–6	Rs 16.30 cr	106.4	Rs 301	Rs 15.07 cr	1.21×
B3	7–9	Rs 4.07 cr	32.8	Rs 228	Rs 11.39 cr	0.91×
B4	10–12	Rs 0 → stack added	17.8	Rs 213	Rs 10.64 cr	0.85×
B5	13–15	Rs 5.74 cr	24.5	Rs 220	Rs 10.97 cr	0.88×
B6	16–18	Rs 3.59 cr	20.2	Rs 215	Rs 10.76 cr	0.86×
B7	19–20	Rs 1.43 cr	16.5	Rs 212	Rs 10.58 cr	0.85×

Table 11: Repricing block schedule. The power component (Rs 171.3/kg) is the same in every block under the constant-price-year assumption. The stack replacement at year 10.6 adds Rs 7.17 cr to the rate base, which then depreciates over the remaining plant life.

NPV-neutrality. Rolling 3-year repricing on a declining rate base is exactly NPV-neutral to both parties: the 20-year PV-levelised price to the offtaker equals the flat 20-year LCOH of Rs 268/kg. The repricing structure distributes the same total value across time without extracting rent from either side. The dashboard asserts this numerically in its self-test, including with additional charges and subsidies applied.

Commercial risk. Under 7-year amortisation the contract price falls from Rs 326 to Rs 228 at the second renewal — a 30% cut visible to the offtaker from day one. The developer should contract the formula (rate base plus indexed power), not the schedule, and disclose the rate-base balance at each reset. The stack replacement causes a price step-up at block 5, which should be pre-disclosed.

12. Independent Arithmetic Checks

The following checks were computed using a second, independent script with no imports from the main model, and verified against the dashboard's data.json. All results must match to within Rs 1.0/kg tolerance.

```
Check 1 - Capital Recovery Factor
  CRF(10%,20) = 0.10 x 1.10^20 / (1.10^20 - 1) = 0.117460    OK
  CRF(10%, 7) = 0.10 x 1.10^7 / (1.10^7 - 1) = 0.205405    OK
  CRF(10%, 3) = 0.402115          CRF(0%,10) = 1/10 = 0.100    OK

Check 2 - Capex Build-Up
  39,000 x 1.59 = 62,010 Rs/kw    OK

Check 3 - Plant Sizing for 500 TPA
  MW      = 500,000 x 52 / 5,654 / 1,000 = 4.5985 MW    OK
  capex   = 4,598.51 kW x Rs 62,010 = Rs 28.5154 cr    OK

Check 4 - LCOH Total (independent calculation)
  landed  = (2.843 + 0.250) x 1.065 = 3.2940 Rs/unit
  cap_rec  = 285,183,990 x 0.117460 / 500,000 = 66.995
  OM       = 285,183,990 x 0.04 / 500,000 = 22.815
  stack    = 26,093,399 x 0.117460 / 500,000 = 6.130
  power    = 52 x 3.2940 = 171.290
  water    = 0.900
  TOTAL    = 268.130    OK

Check 5 - Stack Replacement Timing
  60,000 / 5,654 = 10.6120 yr; inside 20-yr life,
  outside 7-yr recovery    OK

Check 6 - Block 1 Repricing Consistency
  130.887 + 171.290 + 22.815 + 0.900 = 325.892    OK

Check 7 - MW Invariance
  MW = 1, 5, 10, 50 -> LCOH = 268.1200 in every case    OK

Check 8 - Gujarat vs Rajasthan Differential
  Raj landed = (2.843 + 0.365) x 1.088 = 3.4903
  (3.4903 - 3.2940) x 52 = 10.21 Rs/kg    OK
```

12.1 Additional charge and subsidy checks

These verify Section 7. Each is computed independently from the reference design and compared against the model output.

```
Check 9 - Zero-default inertness
  All Section 7 fields = 0 -> LCOH = 268.130, unchanged    OK

Check 10 - Per-unit opex conversion
  0.10 x 52 = 5.200 -> LCOH 273.330    OK

Check 11 - Capital subsidy, O&M; held on gross
  net capex = 285,183,990 - 45,985,100 = 239,198,890
  cap_rec    = 239,198,890 x 0.117460 / 500,000 = 56.185
  OM         = 285,183,990 x 0.04 / 500,000 = 22.815
  (OM unchanged)          LCOH = 257.320    OK
```

Check 12 - Additional capex raises both cap_rec and OM

gross	= 285,183,990 + 22,992,550 = 308,176,540		
cap_rec	= 72.390 (+5.401)		
OM	= 24.652 (+1.839)	LCOH = 275.361	OK

Check 13 - MW invariance with Section 7 fields populated

MW = 1, 5, 10, 50 -> LCOH identical to 4 dp	OK
---	----

Check 14 - Lump-sum exception behaves as documented

Rs 50 lakh/yr at MW = 1 vs MW = 10 -> LCOH differs	OK
--	----

13. What This Tool Does NOT Model

The following cost items and structural features are excluded from the **default** LCOH calculation. Several can now be entered through the additional-charges and subsidy fields described in Section 7; those are marked below. The remainder are structural and cannot be represented in this model at all.

13.1 Excluded costs — enterable via Section 7

Item	Magnitude	How to enter
Electricity duty	Rs 0.05–0.20/unit for industrial HT consumers; Rs 1.3–5.2 mn/yr on 26,000 MWh	Additional opex, Rs/unit power
STU demand charges	Roughly Rs 0.30–0.35/unit at high load factor, or a fixed Rs/kVA/month. May add Rs 15–25/kg — the largest single excluded item	Additional opex, Rs/unit power or Rs/year
GST on power purchases	Depends on input tax credit treatment for the GH2 end use; not universally available	Additional opex, Rs/unit power; reimbursement via production subsidy
Working capital	Daily GDAM margin, roughly 5–10 days of power cost at peak purchase level	Additional opex, Rs/year (carrying cost)
Land and civil lease or acquisition	Excluded from the Rs 62,010/kW stack	Additional capex, Rs/kW, or additional opex Rs/year for a lease
Insurance	Typically 0.3–0.5% of capex per year; Rs 0.85–1.4 cr/yr on the reference design	Additional opex, % of capex/year
Regulatory and statutory costs	Open access application fees, banking charges, compliance	Additional opex, Rs/year
SIGHT Component II	Electrolyser capex support; would reduce the capital recovery line materially if the scheme covers all-in costs	Capital subsidy, Rs/kW

Table 12: Excluded costs that the Section 7 fields can represent. Magnitudes are indicative and not verified for any specific project.

13.2 Structural exclusions — not representable

Item	Note
Storage or battery	The tool models exchange-only dispatch. Adding storage changes both the cost stack and the hour-selection logic fundamentally.
Captive RE plant or PPA	No wind, solar or hybrid plant. Power is sourced entirely from GDAM.
Price escalation	All figures are real-terms, held at the chosen representative year. No inflation, no trending.
States other than Gujarat and Rajasthan	Verified charge stacks exist for these two states only. Formulas for other states are not validated.
Plant degradation	Electrolyser output degradation over stack life, typically 0.5–1.0% per year of rated capacity, is not modelled.
Partial load operation	The model assumes full-load operation whenever the electrolyser runs. Alkaline units can turn down to roughly 20–40% of rated capacity; partial-load economics are excluded.
Carbon accounting	No carbon credit income or green premium pricing.

Item	Note
Grey hydrogen procurement	The tool computes green LCOH only. A blended green-plus-grey supply strategy requires a grey reference price the developer supplies from their own context.

Table 13: Structural exclusions.

13.3 Data limitations

The low-price regime — cheap hours below Rs 1.5 MCP — is approximately 2.5 years old, visible from AY2024-25 onward. In AY2022-23 and AY2023-24 there were essentially no hours below Rs 1.5, the LCOH minimum exceeded Rs 330/kg, and the present design would not have been economically viable. All four years are included in the tool so that the trajectory can be examined directly.

IMPORTANT At the defaults, the LCOH figures in this tool are a lower bound on actual project cost. The excluded items in 13.1 — particularly STU demand charges and electricity duty — can collectively add Rs 20–40/kg or more depending on state and connection voltage. Section 7 provides the mechanism to enter them; it does not supply the numbers.

14. Glossary

Term	Definition
Additional capex	User-entered capital cost in Rs/kW, added to the base stack. Attracts fixed O&M at the same percentage as base capex.
Additional opex	User-entered operating cost, enterable in one of five units and converted to Rs/kg before entering the LCOH.
AY	August Year — the 22 Aug to 21 Aug window used to define each price year.
Bid ceiling	The maximum landed power price above which the developer stops buying GDAM blocks. The number configured in the GDAM order system.
Capex, gross and net	Gross is the full installed cost including additional capex. Net is gross less any capital subsidy. Capital recovery and the rate base use net; fixed O&M uses gross.
Capital subsidy	Support expressed in Rs/kW, deducted from capex at year zero before the CRF is applied.
CF	Capacity Factor — actual operating hours as a fraction of 8,760 hours per year.
CRF	Capital Recovery Factor — the annuity that repays a present value of 1 over N years at discount rate r: $r(1+r)^N / ((1+r)^N - 1)$.
GDAM	Green Day-Ahead Market — the IEX segment for renewable-attributed power, settled at 15-minute block prices.
GH2	Green Hydrogen — hydrogen produced by electrolysis powered by non-fossil electricity.
ISTS	Inter-State Transmission System — charges waived for GH2 open access projects commissioned before 31 December 2030.
LCOH	Levelised Cost of Hydrogen — full lifecycle cost per kg of H2 produced, including capex recovery, O&M, stack replacement, power and water.
MCP	Market Clearing Price — the exchange price at which a GDAM block clears, in Rs/MWh or Rs/kWh.
Mean landed cost	The arithmetic mean of the landed price across all selected blocks. Because the electrolyser runs at full rated load or not at all, every block bought draws the same power and contributes equally to this mean. It is the number that enters the power line of the LCOH — different from, and always below, the bid ceiling.
Min-run block	Minimum continuous operating duration for the electrolyser. Defaults to 2 hours, or 8 blocks of 15 minutes.
MW-invariance	The property that the LCOH-optimal ceiling and CF are independent of plant size, because all cost and output items are proportional to MW. Holds for every input except a lump-sum Rs/year cost.
OA	Open Access — the right to wheel power over the grid between buyer and seller independently.
Power subsidy	Support expressed in Rs per unit of power, converted to Rs/kg by multiplying by SEC.
Production subsidy	Support expressed in Rs per kg of hydrogen, deducted directly from the LCOH. GST reimbursement is entered here.
Rate base	In the repricing module, the unrecovered capital balance on which the annual return is calculated. Opens at net capex.
SEC	Specific Energy Consumption — kWh of electricity required to produce 1 kg of H2. Defaults to 52 kWh/kg for alkaline.

Term	Definition
Sinking fund	CRF × NPV of future stack replacement expenditures, expressed as a levelised annual Rs/kg charge.
Stack replacement	Periodic replacement of the electrolyser cell stack at 60,000 operating hours by default, costing 40% of electrolyser capex.
TPA	Tonnes per annum of hydrogen produced.
Unconstrained MCP	The GDAM price that would have cleared in the absence of network congestion. Used throughout.

Table 14: Glossary of terms used in the dashboard and this document.